

MOHAMED AZIZ CHOUAIBI

Edge AI Engineer

Edge AI engineering student focused on computer vision, embedded deployment, and ADAS systems.

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EDUCATION

National Institute of Applied Sciences and Technology (INSAT), Tunisia2022 – 2027

- Engineering Studies: *Industrial Computing and Automation (IIA)*.

- Specialization: **Artificial Intelligence, Embedded Systems, and Computer Vision.**

PROFESSIONAL EXPERIENCE

GENOW — Edge AI Engineer (Part-time)Jul 2025 – Present

- Spearheaded the design and deployment of a YOLO-based perception pipeline combining detection and semantic segmentation for the HawkVision ADAS system on NVIDIA Jetson Nano.
- Drove lightweight model optimization and hardware-aware benchmarking, identifying accuracy-latency trade-offs and accelerating edge deployment via ONNX and TensorRT quantization.
- Improved real-time inference robustness on constrained edge hardware through iterative training, data augmentation, and post-processing with Python and fine-tuned Vision-Language Models, contributing 3–6 mAP points of improvement across 15,000+ labeled images and reducing failure cases in low-visibility scenes.
- Initiated an AI-assisted data cleaning pipeline and a rule-based vehicle trajectory prediction module to strengthen downstream ADAS decision-making on embedded systems.
- Keywords:** Edge AI, Embedded AI, TensorRT, ONNX, NVIDIA Jetson, YOLO, Object Detection, Segmentation, ADAS, Vision-Language Models, Trajectory Prediction

PERSONAL PROJECTS

AI-Based Energy Management System for Hybrid Electric Vehicles2026 – Ongoing

- Launched an AI-based energy management system for hybrid electric vehicles in CARLA Simulator, modeling 20+ driving-state variables to optimize engine-motor power split and targeting 8–12% fuel-efficiency gains with lower emissions through baseline policy evaluation.

Deep Learning for Retinal Disease Detection — Deep Learning, Computer Vision, PyTorch[GitHub](#)2025

- Implemented a PyTorch classifier for diabetic retinopathy stage classification with a ResNet-152 backbone, reaching 85% accuracy on 224×224 retinal fundus images.

Smart Factory Automation — PLC Programming, TIA Portal, Factory I/O[GitHub](#)2025

- Led the programming and validation of Siemens S7-1500 PLC control logic to synchronize a virtual smart production line in Factory I/O.

RFID Crowd Management (Research Paper Prototype)[Research Link](#)2024

- Proposed and implemented an RFID-based tracking system for intelligent crowd management at Al-Rawdah Al-Sharifah as part of a research prototype with Shaqra University.

SKILLS

- Programming Languages:** Python, C, C++
- Frameworks & Libraries:** PyTorch, TensorFlow, OpenCV, FastAPI, Hugging Face, YOLO, ROS 2
- Embedded / Edge:** NVIDIA Jetson Nano, STM32, ESP32, RFID, RTOS basics, ONNX, ADAS basics
- Machine Learning:** Supervised Learning, Linear Regression, Logistic Regression, Classification
- Deep Learning:** CNNs, RNNs, Transformers, LLMs, VLMs, Transfer Learning, Fine-Tuning
- Automation & Industrial Control:** TIA Portal, PLC Programming
- MLOps & Tools:** Linux, Git, GitHub, DVC, MLflow, Docker
- Focus Areas:** Computer Vision, Embedded AI, ADAS, Industrial Automation
- Languages:** Arabic (Native), English , French

CERTIFICATIONS

- Deep Learning Track — DataCamp[Credential](#) (2025)
- Machine Learning Engineer — DataCamp[Credential](#) (2026)
- Supervised Machine Learning: Regression and Classification — Coursera[Credential](#) (2025)